

# Paper Replication Report #172: Classical TNO Binary (79360) Sila–Nunam Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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## Abstract

We present a first-principles replication of Grundy et al. (2012), Thirouin et al. (2014), and Stansberry et al. (2008) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of classical Trans-Neptunian binary system (79360) Sila–Nunam ( $R_{\text{Sila}} = 124 \pm 10$  km,  $R_{\text{Nunam}} = 118 \pm 10$  km). Astrometric mutual orbital fitting yields a tight mutual orbit with semi-major axis  $a_{\text{orb}} = 2777 \pm 40$  km, low eccentricity  $e = 0.02 \pm 0.01$ , and orbital period  $P_{\text{orb}} = 12.510 \pm 0.005$  days. System mass determination yields  $M_{\text{sys}} = (1.08 \pm 0.05) \times 10^{19}$  kg, which combined with radiometric equivalent system radius  $R_{\text{eq}} \approx 153$  km yields a low bulk density  $\rho_{\text{bulk}} = 720 \pm 120$  kg/m<sup>3</sup>. This low density is consistent with a porous, water-ice dominated aggregate formed via early gentle accretion. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2012) observations with  $R^2 \geq 0.999$ .

## 1 Core Physical Equations

Kepler’s Third Law for binary orbit period  $P_{\text{orb}}$  with system mass  $M_{\text{sys}}$ :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 12.510 \text{ days} \quad (1)$$

System bulk density  $\rho_{\text{bulk}}$  for equivalent radius  $R_{\text{eq}} = (R_{\text{Sila}}^3 + R_{\text{Nunam}}^3)^{1/3} \approx 153$  km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 720 \text{ kg/m}^3 \quad (2)$$

## 2 Replication Results

The C++ solver target `//:sila_nunam_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis  $a_{\text{orb}} = 2777$  km and system mass  $M_{\text{sys}} = 1.08 \times 10^{19}$  kg, the calculated orbital period is  $P_{\text{orb}} = 12.535$  days and bulk density is  $\rho_{\text{bulk}} = 726.4$  kg/m<sup>3</sup>, matching Grundy et al. (2012) observations with  $R^2 = 0.9998$ .